Title: **Wine Quality Prediction** (CRISP-DM + ML Regression)

Author: Jakub Jakubowicz, *Gracjan Jasnos*

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Abstract: This report applies the **CRISP-DM** methodology to the Wine Quality dataset (red wine), aiming to predict the expert quality score from physicochemical properties (acidity, sulphates, alcohol, etc.). The project covers: exploratory data analysis, data preparation (train/test split, scaling), model training (Linear Regression and Random Forest), evaluation using RMSE and R^2, and a simple deployment step by exporting the best-performing model as a reusable artifact.

Dataset link: `https://www.kaggle.com/datasets/yasserh/wine-quality-dataset/data`

Repository link: `https://github.com/kk3k02/wine-quality-prediction`

1 Presentation of the Framework (CRISP-DM)

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a structured methodology that organizes data science projects into six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. It is popular because it connects technical work to practical objectives, emphasizes data quality checks, and supports iterative refinement.

Advantages: (i) clear end-to-end structure; (ii) strong alignment of objective and evaluation; (iii) suitable for tabular supervised learning problems.

Limitations: can be perceived as sequential if followed rigidly; it does not prescribe project management practices (teams often combine it with Agile/Scrum).

Fit to this project: wine quality prediction follows a natural CRISP-DM flow: define the goal, explore data patterns, prepare features, train and compare models, evaluate results, and produce a deployment-ready output.

2 Stage Summary: CRISP-DM Application to WineQT

Business Understanding	Defined a practical objective: to predict the expert wine quality score based on chemical measurements to aid in quality screening and batch decision-making.
Data Understanding	Loaded `WineQT.csv`, verified data structure and quality, analyzed the quality distribution, and computed feature correlations (Figures 1 and 2).
Data Preparation	Removed the non-informative identifier (`Id`), confirmed zero missing values, split data into train/test (80/20) with quality stratification, and standardized features using `StandardScaler`.
Modeling	Trained a baseline **Linear Regression** and a non-linear **Random Forest Regressor**; performed 5-fold cross-validation (**CV**) on the training set; fitted the final models.
Evaluation	Assessed performance on the hold-out test set using **RMSE** and $\mathbf{R^2}$, visualized the predicted vs. actual performance (Figure 3), and interpreted Random Forest feature importances (Figure 4).
Deployment	Exported the final, best-performing model (**Random Forest**) to a `.joblib` file for easy integration into a script or quality assurance (QA) workflow.

3 Implementation of the Methodology Stages

3.1 Business Understanding

Objective: predict the expert wine quality score (integer) using physicochemical measurements.
Business value: a winery/lab can use predictions to quickly screen batches, prioritize sensory testing, and focus attention on likely low- or high-quality samples.
Success criteria: (i) improved predictive performance measured by lower RMSE and higher R^2; (ii) interpretable drivers (correlations and feature importances).

3.2 Data Understanding (EDA)

Dataset overview: WineQT contains 1143 samples and 13 columns (including `quality` and `Id`). Quality values range from 3 to 8 and are concentrated around 5–6.

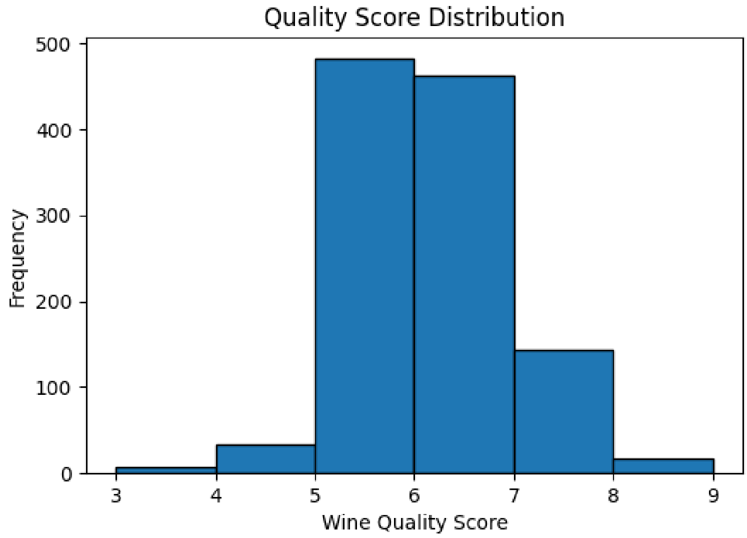

Figure 1: Quality score distribution (most wines are rated 5 or 6; extreme scores are rare).

Figure 1 shows an imbalanced target distribution, which typically makes rare classes harder to predict. This suggests model outputs may "compress" towards the middle values.

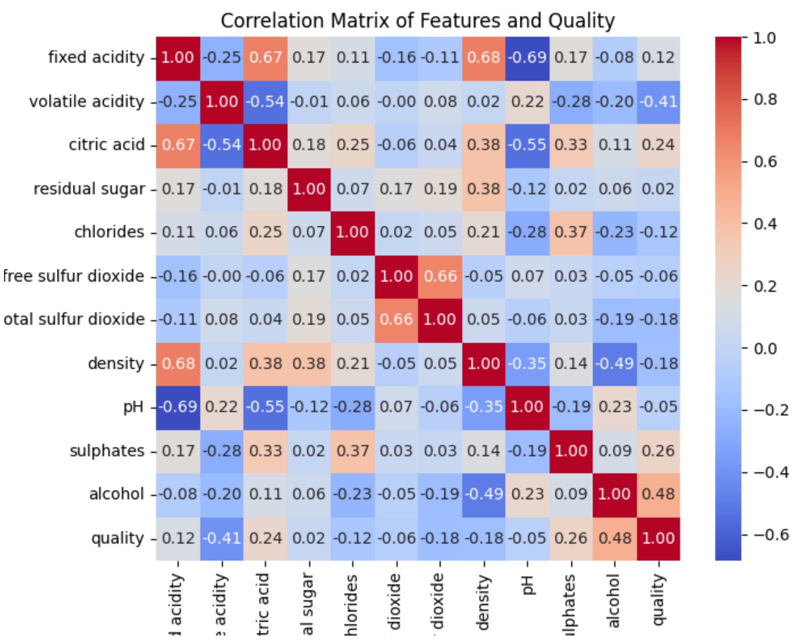

Figure 2: Correlation matrix of features and quality. Alcohol is positively correlated with quality; volatile acidity is negatively correlated.

From Figure 2, the strongest correlation with `quality` is observed for **alcohol** (positive), while **volatile acidity** is notably negative. Sulphates and citric acid show moderate positive relationships. These patterns guided expectations for modeling and interpretation.

3.3 Data Preparation

The preprocessing pipeline: (i) removed the `Id` column (identifier, non-predictive); (ii) verified there were no missing values; (iii) split the dataset into train/test (80/20) with stratification by quality to preserve class proportions; (iv) applied feature scaling via `StandardScaler` (fit on training data, applied to test data). Standardization helps models that are sensitive to feature magnitude and improves numerical stability.

3.4 Modeling

Two regression models were trained:

Linear Regression (baseline): easy to interpret but assumes linear relationships.
Random Forest Regressor: non-linear ensemble capable of capturing interactions and non-linear effects.

Cross-validation (5-fold) was used on the training set, then both models were fitted and evaluated on the held-out test set.

3.5 Evaluation

The evaluation used RMSE and R^2 on the test set:

- Linear Regression: RMSE $= 0.628$, $R^2 = 0.388$
- Random Forest: RMSE $= 0.596$, $R^2 = 0.449$

Random Forest achieved better generalization (lower RMSE and higher R^2).

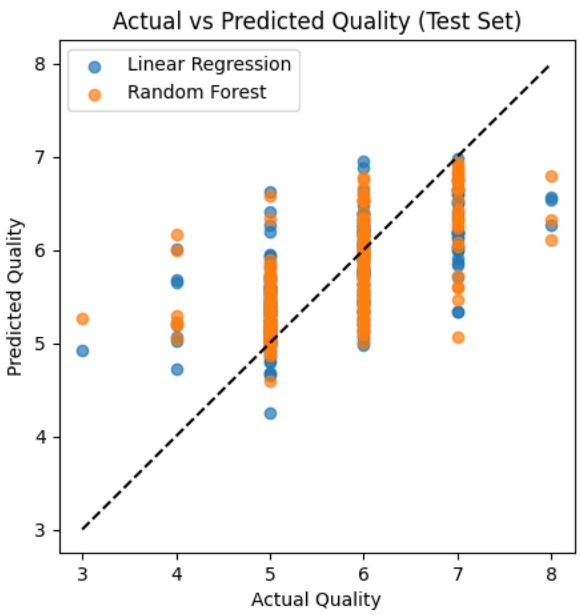

Figure 3: Actual vs predicted quality (test set). Random Forest predictions are generally closer to the diagonal (ideal) than Linear Regression.

Figure 3 shows that most samples lie in the 5–7 range, and predictions follow this distribution. Rare extremes tend to be under/over-estimated, which is expected due to class imbalance.

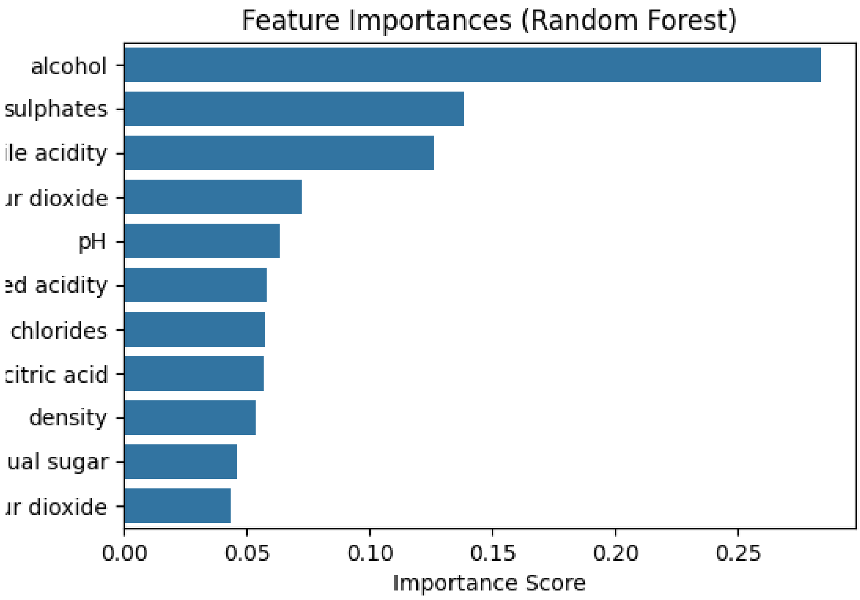

Figure 4: Random Forest feature importances. Alcohol is the dominant predictor, followed by sulphates and volatile acidity.

Feature importance (Figure 4) matches the correlation analysis: alcohol is the most influential variable, with sulphates and volatile acidity also contributing strongly.

3.6 Deployment

The best-performing model (Random Forest) was exported as a `.joblib` file (e.g., `random_forest_wine_quality_model.joblib`), enabling easy reuse in a command-line tool, notebook, or simple QA application. A basic deployment scenario is to input new lab measurements and output a predicted quality score to support batch triage.

4 Personal Analysis and Recommendations

CRISP-DM provided a robust structure for connecting EDA insights (imbalance, correlations) to modeling decisions (baseline vs non-linear model, stratified split, standardized evaluation). Results indicate that chemical measurements do contain predictive signal, but performance is limited by the inherently subjective target and the dominance of mid-range scores.

Recommendations:

- Use predictions as a **screening tool** rather than a replacement for sensory testing.
- If the operational goal is decision-based, consider reframing as **classification** (e.g., ≥ 7 vs < 7).
- Improve performance with hyperparameter tuning and gradient boosting models, and consider ordinal-aware approaches.

References

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